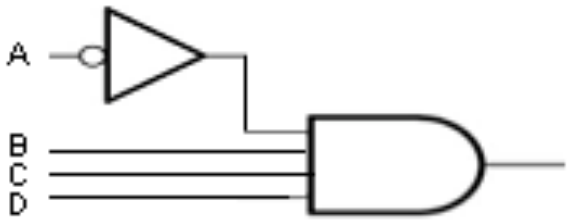
PART A: 20 OBJECTIVE QUESTIONS [20 MARKS]

INSTRUCTION: Answer PART A in the OBJECTIVE ANSWER SHEET given in PAGE 6.

1. Which of the following is NOT a type of Adder?
   A. Half
   B. Full
   C. Regulator
   D. Parallel

2. Which of the following functions can be used to detect whether your bank account balance is ZERO?
   A. Parity generator
   B. Multiplexer
   C. Comparator
   D. Encoder

3. When the output is active-HIGH for the decoding gate shown below, what is the binary code appearing on the inputs? The MSB is A.



   A. 1110
   B. 0111
   C. 1100
   D. 1000

4. _____ reduces the number of bits needed to represent given information.
   A. Parity Checker
   B. Parity Generator
   C. Decoder
   D. Encoder

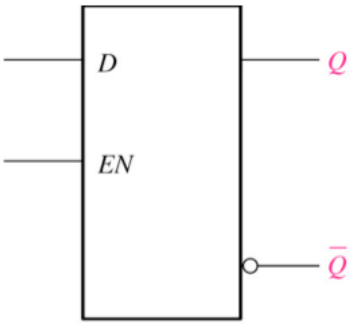
5. MUX is also known as _____.

A. data distributor

B. data selector

C. adder

D. decoder



PART B: SUBJECTIVE QUESTIONS [80 MARKS]

QUESTION 1 [15 MARKS]

a) Two binary numbers are given as, A = $\{A_2A_1A_0\} = 011_2$ and B = $\{B_2B_1B_0\} = 110_2$.

i. Design a parallel adder using Full Adder. Label the diagram with A_0, A_1, A_2, B_0, B_1, B_2, $\sum_0$, $\sum_1$, $\sum_2$, C_{in}, C_{out}, MSB, LSB and GND (ground). [4m]

ii. Perform the arithmetic addition A + B, and label all the values in (i). [3m]

iii. Based on how Full Adder handles the carry bit, list TWO types of parallel adder. [2m]

b) Figure 1 shows the symbol of one of combinational logic functions.

i. Name the combinational logic function X and describe its purpose. [2m]

ii. Reproduce Table 1 in the answer booklet and fill it up with the correct answers. [4m]

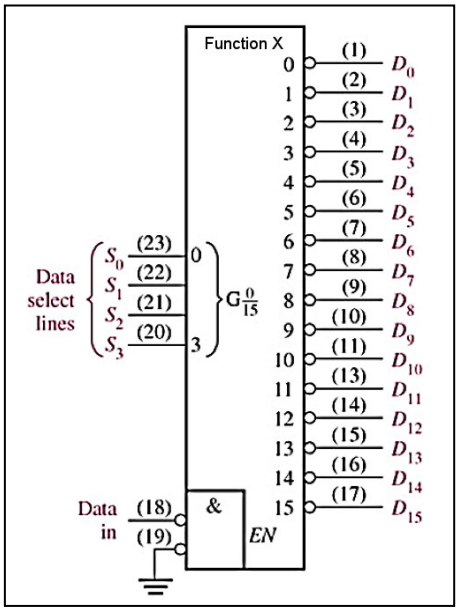


Figure 1: The symbol of combinational logic function X

Table 1

Data In	**S_3**	**S_2**	**S_1**	**S_0**	**Selected Output Pin**	**Output Signal Level**
H	H	H	L	H		
L					D_5	
					D_9	H
	L	H	L	L		L

Note: H = High = 1, L = Low = 0.